

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q.1 Solve Any Two of the following.

- | | | |
|--|---------------|---|
| A) Explain different parts of a Java program with an appropriate example | (Application) | 6 |
| B) What are the advantages of the object-oriented programming language? | (Understand) | 6 |
| C) List the primitive data types available in Java and explain. | (Understand) | 6 |

Q.2 Solve Any Two of the following.

- | | | |
|---|---------------|---|
| A) What are the benefits of break and continue statements explain with example? | (Analysis) | 6 |
| B) Discuss default constructor and parameterized constructor with the help of an example in Java? | (Application) | 6 |
| C) How do we implement polymorphism in JAVA? Explain briefly | (Understand) | 6 |

Q.3 Solve Any Two of the following.

- | | | |
|--|----------------|---|
| A) a) Write a program to find the smallest and largest number from given input array?
For example : Input: arr[]={ 13,4,2,5,9}
Output : Maximum is 13 and Minimum is 2 | (Application)) | 6 |
| B) Write a program to check whether enter number is palindrome or not? | (Application) | 6 |
| C) Write the differences between interface and abstract class | (Understand) | 6 |

Q.4 Solve Any Two of the following.

- | | | |
|---|---------------|---|
| A) Define inheritance. What are the benefits of inheritance? How to prevent a class from inheritance? | (Understand) | 6 |
| B) Write a program to demonstrate hierarchical and multiple inheritance using interfaces | (Understand) | 6 |
| C) Define polymorphism. Explain run time polymorphism with the help of example. | (Application) | 6 |

Q.5 Solve Any two of the following.

- | | | |
|--|---------------|---|
| A) What is the difference between error and an exception? Explain with help of example | (Understand) | 6 |
| B) How to create a user defined exception? | (Application) | 6 |

- C) Write an html page and also provide a JavaScript for accepting a user ID and password from the user to ensure that the input is not empty (Application) **6**
- *** End *****